

Car Rental Use Case

PROJECT OVERVIEW

A structured data pipeline designed to clean, standardize, and integrate multi-city car rental data from sources such as reservations, telematics, maintenance logs, and payment systems, enabling accurate analytics and informed business decision-making.

Presented By
Ashutosh Jha
B.Tech CSE (Final Year)
KIIT University

Capgemini Sprint Project
Data Engineering & Analytics

Problem Statement

Real-world car rental data contains inconsistencies, format variations, and duplicate records, making it unreliable for accurate analytics and business decisions.

Data Inconsistency

- Inconsistent Vehicle IDs
- Invalid & mixed timestamps
- Odometer format variations
- Fuel level inconsistencies

Operational Impact

- Incorrect pricing calculations
- Missed maintenance schedules
- Inaccurate vehicle availability
- Poor customer experience

Business Risk

- Revenue loss due to errors
- Fraud and compliance risks
- Billing mismatches
- Customer trust issues



Dataset overview

Segment of Raw Data

Vehicle_ID	Pickup_TS	Odo_Start	Rate	City
CAR-188	2025-01-22T00:38	30434 km	\$20/day	MUM
CAR_029	2025/02/04 13:33	24285 km	\$20/day	Delhi
CAR_006	2025-01-10T23:38	79,142	Rs1500/day	chennai
CAR411	02/01/2025 13:52	69,349	Rs1500/day	MUM
CAR-327	01/02/2025 02:35	57,384	₹1,800/day	delhi
car-295	04/06/2025 16:29	32076	1800 perday	delhi
car342	2025-06-23 10:25	26291 km	NaN	Mumbai

Segment of Cleaned Data

Customer_ID	Pickup_TS	Odo_Start	Rate	City	Payment
CUST-1176	2025-03-08 03:38:00	51887.0	1840.00	Delhi	UPI
CUST-0561	2025-03-28 17:23:00	43960.0	1800.00	Delhi	CARD
CUST-1389	2025-04-27 11:23:00	37826.0	2500.00	Delhi	CARD
CUST-0580	2025-05-17 02:58:00	60155.0	2300.00	Chennai	UPI
CUST-0298	2025-06-06 19:17:00	24553.0	1835.27	Delhi	CARD
CUST-1129	2025-06-11 20:44:00	35500.0	1500.00	Chennai	CARD

PROVENANCE & SIZE

~5000

Total Rows

~21

Total Columns

~473

Duplicated Rows
(Intentional Noise)

Data Characteristics

- Contains missing and inconsistent values
- Includes duplicate and overlapping bookings
- Mixed timestamp formats (real-world issue)
- Multi-city and multi-vehicle data
- Designed to test data cleaning & validation logic

Major Issues

Inconsistent Vehicle IDs

Invalid Rental Duration

Odometer Unit Inconsistency

Fuel Level Format Variations

Duplicate Reservation Records

*Invalid & Mixed Timestamp
Formats*

Rental Rate Format Differences

Mileage Validation Issues



Tools & Libraries Used



Core programming language used to build the ETL data cleaning process, implement transformation logic, and process large CSV datasets efficiently.



Supports high-performance numerical computations and vectorized operations for efficient data processing.



Used for data ingestion, manipulation, and transformation including duplicate removal, timestamp normalization, unit standardization, and dataset restructuring.



Interactive environment used for developing, testing, and documenting the data cleaning



Used to create visualizations and exploratory analysis plots to understand patterns, anomalies, and data quality issues.



Provides advanced statistical visualizations to analyze relationships, distributions, and trends within the rental dataset.



Used for location-based processing and city normalization, enabling geographic data validation and enrichment.



Used for automated testing of cleaning and transformation steps to ensure reliability and correctness of the process.

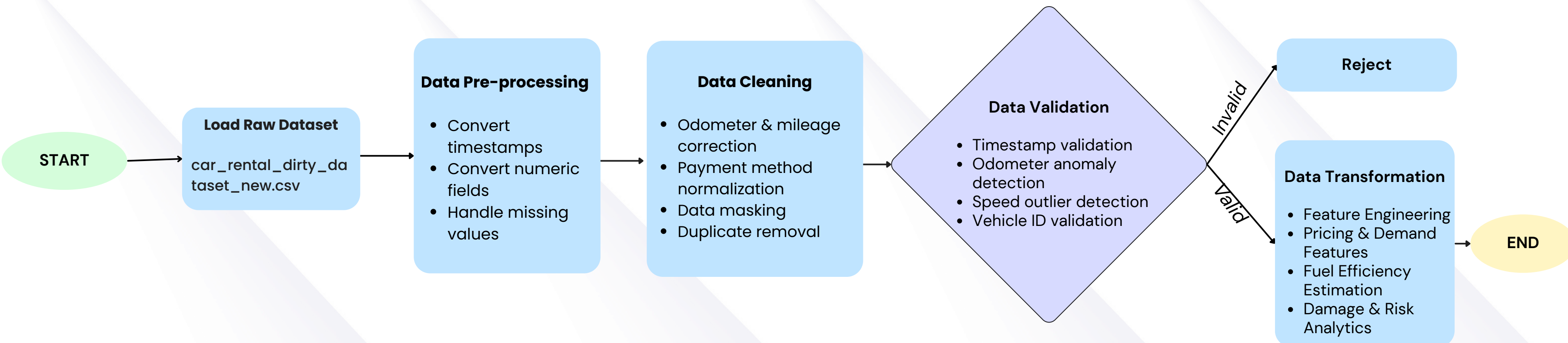
Implementation Overview

We built an end-to-end Car Rental Data Cleaning & Transformation workflow in Python to process raw CSV rental data into a clean, analysis-ready dataset.

Additional standardization was applied to ensure uniform formats for Vehicle IDs and city names, along with normalization of fuel levels for accurate analysis.

Data validation ensures consistency and reliability by identifying issues before further analysis.

Key checks: Timestamp consistency, odometer unit standardization, overlap detection, and fuel level validation.



Thank You

I appreciate your time and attention.
I am happy to answer any questions.

Ashutosh Jha
B.Tech CSE | KIIT University



Key Outcomes

- Built an end-to-end data pipeline
- Improved data quality and consistency
- Enabled reliable analytics
- Reduced errors and data inconsistencies